

Sid Diamond

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Profile

Recently completed MSci in Physics at Imperial College London, with a focus on uncertainty quantification in both transformer-based NLP systems and applied graph theory algorithms for modelling complex systems. I also have software engineering experience leading development of an MVP multi-agent planning application. My recent research has also covered dataset bias in supervised learning, information theory, an independent project with Imperial's High Performance Computing group applying transformer-based NLP to policy documents, and freelance research using information theory to study the relationship between AI systems and their internal models of the world.

Education

Integrated MSci in Physics 2022–2026

Imperial College London, UK

Overall: 2:1 (65%) — Master's Year: 2:1 (69%)

Relevant Modules:

- Object-Oriented Programming (98.70%)
- Information Theory (91.00%)
- Statistics (72.90%)
- Problem Solving (79.30%)
- Master's thesis on predictive modelling of growing complex system networks (74.15%)
- Undergraduate thesis on uncertainty quantification in transformer-based NLP (69.50%)
- Computational Sociology: dataset bias in supervised machine learning (67.00%)
- Communicating Physics: educational policy report (84.10%, highest mark in cohort)

A-Levels 2014–2021

Stoke Newington School and Sixth Form, London, UK

Further Maths (A*), Maths (A*), Physics (A*), Chemistry (A*), EPQ (A*)

Awards: Chemistry Olympiad Silver (2021); Kelvin Prize Winner (2020, *Peterhouse College Cambridge University Science Competition*)

Research Experience

Master's Thesis: Network Growth in Finite-Sized Networks 2025–2026

Theoretical Physics Group, Imperial College London — Supervisor: Associate Professor Tim Evans

- Built and simulated a predictive model for connectivity growth in finite complex systems (such as growing neural networks or AI agent networks), with emphasis on empirical validation and model behaviour under realistic constraints (the effects of homophily and cumulative attachment)
- Created a statistical goodness-of-fit test to quantify confidence in the predictive model and assess agreement with observed network behaviour
- Produced research code and write-up that were validated by my supervisor and an external marking panel as a meaningful extension of existing literature; work is currently being used by visiting researchers
- Code and report: Master's thesis code and write-up

Hate Speech Detection Dataset Research 2025–2026

Advanced Humanities Department, Imperial College London — Supervisor: Dr Catherine Webb

- Conducted empirical research on how benchmark datasets can introduce bias into supervised machine learning systems for hate speech detection, with a focus on evaluation quality and downstream fairness implications
- Analysed limitations of benchmark-driven model assessment and explored how dataset design can distort conclusions about model performance in real moderation settings
- Proposed a policy framework to mitigate harms arising from biased datasets and evaluation pipelines
- Report: On Hate Speech Detection

Uncertainty in Transformer-Based NLP Algorithms 2024

Independent Research & Undergraduate Thesis — Supervisor: Dr Aidan Crilly

- Worked with the Imperial High Performance Computing research group to develop a transformer-based NLP tool for topic extraction and document-level scoring of actionable policy insight in economic climate policy documents

- Implemented BERT-based modelling workflows and analysed model outputs for applied NLP tasks on real-world policy text. Achieved near-industry standard f1 scores across range of datasets
- Extended the project into an undergraduate thesis on uncertainty quantification in transformers, investigating per-parameter variance estimates due to fine-tuning, overall model uncertainty using MonteCarlo dropout, and information loss associated with dimensionality reduction. Only self-proposed project in cohort
- Code and write-up: Transformer Uncertainties in NLP

Educational Policy Report

2023

Communicating Physics Coursework, Imperial College London

- Authored an evidence-based policy report combining quantitative attainment analysis with qualitative classroom observation
- Awarded the highest mark in the cohort, demonstrating clear technical communication for non-specialist and policy audiences
- Report: Educational Policy Report

Professional Experience

AI Safety Researcher

2026

Dovetail Research — Supervisors: MATS Scholar Alex Altair and Dr Alfred Harwood

- Conducted paid freelance research for Dovetail Research, an ARIA-funded nonprofit researching the mathematical foundations of AI safety
- Investigated extensions of the Touchette–Lloyd (2003) theorem to better characterise mutual information between an agentic AI system’s world model and the external world
- Contributed to theoretical work aimed at supporting more empirical measures of AI goal alignment
- Report: Agentic Foundations AI Research

Product Engineer Intern

2025

Mindlace Technologies

- Led development of an MVP agentic planning application, implementing full-stack product functionality in a fast-moving startup environment
- Built practical workflows for an AI-driven planning product, combining rapid prototyping with modern software tools and iterative product development
- Gained hands-on experience translating product ideas into working systems for a consumer-facing agentic AI application
- Project link: Agentic AI Planning Application MVP

Science & Maths Tutor

2021–2024

Self-Employed

- Taught A-level Physics and mathematics across a range of student ability levels, including a four-month placement teaching 25 students
- Developed strong communication skills through explaining technical concepts clearly, adapting explanations to diverse learner needs, and building confidence in quantitative reasoning

Technical Skills

Programming: Python, TypeScript, NetLogo

ML / NLP: PyTorch, HuggingFace Transformers, BERT, NLP pipelines, model uncertainty analysis, statistical evaluation, supervised learning, unsupervised learning, data analysis

Data & Scientific Computing: NumPy, SciPy, Pandas, Matplotlib, Plotly, statistical modelling, dimensionality reduction, network analysis, graph theory, linear algebra, fine-tuning

Software & Prototyping: GitHub, LaTeX, Supabase, n8n, Shadcn/ui, Pinecone, OpenRouter, Firecrawl, Tavily, OpenAPI, AWS cloud, Microsoft Azure

Additional Activities

Sport: Imperial Men’s Football 1st and 2nd teams; previously captain of the 2nd team. Also enjoy recreational running.

Journalism: Comment writer for *Felix* (Imperial College student newspaper), including articles on ESG and artificial intelligence.

Music: Guitar.

References available upon request